EduAirControl: Project Annexes

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Purpose

This folder gathers the supporting material behind the main deliverables: API contracts, architecture decisions, infrastructure files, risks, screenshots and the test inventory. They are unmodified copies of the project files, except for Annexes D and G, which were written for this delivery. Annexes D and G and this index exist in Spanish (_ES) and English (_EN).

Contents

• | Annex | Content | Location |

• |---|---|---|

• | A | OpenAPI 3 contracts of the services (8 files, including the shared component) | Anexo_A_Contratos_OpenAPI/ |

• | B | 17 architecture decision records (ADR-001 to ADR-017) | Anexo_B_Decisiones_Arquitectura/ |

• | C | Docker Compose, wait and smoke-test scripts, Kubernetes manifests | Anexo_C_Infraestructura/ |

• | D | Project risk register (summary) | Anexo_D_Registro_Riesgos_ES and Annex_D_Risk_Register_EN (.pdf and .md) |

• | E | 39 screenshots used in the manuals | Anexo_E_Capturas_Pantalla/ |

• | F | Inventory of the 285 automated tests | Anexo_F_Inventario_Pruebas/ |

• | G | Glossary | Anexo_G_Glosario_ES and Annex_G_Glossary_EN (.pdf and .md) |

Annex A. OpenAPI contracts

• | File | Service | Documented paths |

• |---|---|---|

• | api-gateway.yaml | API Gateway | 2 |

• | auth-service.yaml | ms-security | 12 |

• | ms-classroom-management.yaml | Environments and campuses | 7 |

• | ms-environment-monitoring.yaml | Monitoring and analysis | 10 |

• | ms-sensor-management.yaml | Sensors and devices | 10 |

• | ms-user-experience.yaml | Favorites and preferences | 8 |

• | ms-user-management.yaml | User profile | 4 |

• | _shared.yaml | Shared components | – |

Annex B. Architecture decisions

The ADR files themselves are written in English.

• | ADR | Topic |

• |---|---|

• | 001 | Documentation language |

• | 002 | Bounded contexts and microservice boundaries |

• | 003 | Database strategy |

• | 004 | Communication patterns |

• | 005 | API Gateway |

• | 006 | Authentication and authorization (token flow partially superseded by 017) |

• | 007 | Event delivery and consistency |

• | 008 | Database migration tool (Liquibase) |

• | 009 | Internal hexagonal architecture |

• | 010 | Containerization strategy |

• | 011 | Production deployment strategy |

• | 012 | Observability stack |

• | 013 | Circuit breaker adoption |

• | 014 | Environmental sensor variables and thresholds |

• | 015 | Folding alerts into ms-environment-monitoring |

• | 016 | Multi-tenancy by institution and campus |

• | 017 | OAuth2/OIDC authorization server and Backend-for-Frontend |

Annex C. Infrastructure

• docker-compose.yml: orchestrates the services, PostgreSQL, Redis and RabbitMQ locally.

• wait-for-services.sh and smoke.sh: wait until the services are healthy and run an end-to-end check (register, login, logout and reuse of a revoked token).

• k8s/: namespace, configmap, ingress, PostgreSQL, RabbitMQ, Redis, applications and a secrets template (secrets.example.yaml).

Warning. These files contain sample development values only (for example REPLACE_ME or change-me-in-vault). No .env files or keys were copied. In a real environment, secrets must come from a vault and must not be repeated.

Annex E. Screenshots

The images image.png to image-38.png come from the technical, installation, administrator and user manuals. They keep their original names so the manuals can still link to them.

Annex F. Test inventory

Inventario_Pruebas_Automatizadas.csv lists every test method by service and class, its type and whether it needs Docker. The result column is NOT RUN: it is updated with the real execution (see the 07_Pruebas folder).

Cited material that was not in the zip

The ESP32 firmware files (esp32_sensor_sender.ino and infra/iot/esp32-node, mentioned in the risk register) and the iot/esp32/ sensor-test folder were not in the zip received. If they exist in the repository, they should be copied to an Anexo_H_Firmware_IoT/ folder.